

Project Title: Basic File Handling in Python

Description:

The Basic File Handling project demonstrates how to work with text files in Python. The program allows users to **read data from a file**, **search and replace specific words**, and then **save the modified content back** into the same file (or a new file). It also includes **error handling mechanisms** to manage scenarios such as missing files, permission errors, or invalid input, ensuring the program runs smoothly in real-world situations.

Key Features:

- Read text data from a file
- Find and replace specific words in the file content
- Save updated content back into the file
- Error handling for file-related exceptions (e.g., file not found, permission issues)
- User-friendly prompts for input and feedback.

Skills Demonstrated:

- **File Input/Output:** Using `open()`, `read()`, `write()`, and context managers (with)
- **String Manipulation:** Finding, replacing, and updating words in text
- **Exception Handling:** Handling errors with try-except blocks
- **Practical Python Usage:** Applying basic programming concepts to real-world file operations.

Applications:

This project is useful as a foundation for building more advanced text-processing tools such as:

- Log file analyzers
- Text editors
- Automated report generators
- Data cleaning/preprocessing scripts.

Python code:

```
def find_and_replace(file_name, word_to_find, word_to_replace):

    try:

        # Read file content

        with open(file_name, "r", encoding="utf-8") as file:

            content = file.read()

        # Perform replacement

        if word_to_find in content:

            updated_content = content.replace(word_to_find, word_to_replace)

            # Save changes back to the file

            with open(file_name, "w", encoding="utf-8") as file:

                file.write(updated_content)

            print(f"✅ All occurrences of '{word_to_find}' have been replaced with '{word_to_replace}'.")

        else:

            print(f"⚠️ The word '{word_to_find}' was not found in the file.")

    except FileNotFoundError:

        print(f"❌ Error: The file does not exist. Please check the file name.")

    except PermissionError:

        print(f"❌ Error: You don't have permission to access this file.")

    except Exception as e:

        print(f"⚠️ An unexpected error occurred: {e}")
```

```
def main():
```

```
    print("📁 Basic File Handling Program")
```

```
    file_name = input("Enter the file name (e.g., data.txt): ").strip()
```

```
    word_to_find = input("Enter the word you want to find: ").strip()
```

```
    word_to_replace = input("Enter the word you want to replace it with: ").strip()
```

```
    find_and_replace(file_name, word_to_find, word_to_replace)
```

```
if __name__ == "__main__":
```

```
    main()
```

Output:

📁 Basic File Handling Program

Enter the file name (e.g., data.txt): read

Enter the word you want to find: r

Enter the word you want to replace it with: raw

❌ Error: The file does not exist. Please check the file name.